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SEMICONDUCTOR PACKAGE AND METHOD OF FORMING SAME

Abstract

In an embodiment, a method includes attaching a first package component to a first carrier, the first package component comprising: an aluminum pad disposed adjacent to a substrate; a sacrificial pad disposed adjacent to the substrate, the sacrificial pad comprising a major surface opposite the substrate, a protrusion of the sacrificial pad extending from the major surface; and a dielectric bond layer disposed around the aluminum pad and the sacrificial pad; attaching a second carrier to the first package component and the first carrier, the first package component being interposed between the first carrier and the second carrier; removing the first carrier; planarizing the dielectric bond layer to comprise a top surface being coplanar with the protrusion; and etching a portion of the protrusion.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a continuation of U.S. patent application Ser. No. 18/631,900, filed on Apr. 10, 2024, entitled “Semiconductor Package and Method of Forming Same,” which is a divisional of U.S. patent application Ser. No. 17/678,774, filed on Feb. 23, 2022, entitled “Semiconductor Package and Method of Forming Same,” now U.S. Pat. No. 11,984,422, issued on May 14, 2024, which claims the benefit of the following provisionally filed U.S. Patent application: Application No. 63/230,117, filed on Aug. 6, 2021, and entitled “Innovative SoIC Integration Flow for Robust Reliability Performances,” which applications are hereby incorporated herein by reference.

BACKGROUND

[0002] The semiconductor industry has experienced rapid growth due to ongoing improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, improvement in integration density has resulted from iterative reduction of minimum feature size, which allows more components to be integrated into a given area. As the demand for shrinking electronic devices has grown, a need for smaller and more creative packaging techniques of semiconductor dies has emerged. An example of such packaging systems is Package-on-Package (PoP) technology. In a PoP device, a top semiconductor package is stacked on top of a bottom semiconductor package to provide a high level of integration and component density. PoP technology generally enables production of semiconductor devices with enhanced functionalities and small footprints on a printed circuit board (PCB).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1-5, 6A, 6B, 7A, 7B, and 8-12 illustrate cross-sectional views of intermediate stages in the formation of a semiconductor package in accordance with some embodiments.

DETAILED DESCRIPTION

[0005] The following disclosure provides many different embodiments, or examples, for

implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0006] Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0007] A semiconductor package and the method of forming the same are provided. In accordance with some embodiments of the present disclosure, a first package component is formed in a wafer and comprises an integrated circuit, through-silicon vias, and aluminum pads. Some of the aluminum pads that are electrically connected to the integrated circuit are used for circuit probe testing to determine known good dies (KGDs) within the wafer. A recess and a protrusion may be formed in the aluminum pads as a result of the circuit probe testing. After encapsulating the aluminum pads in a dielectric bond layer, the KGDs may be singulated, flipped, and attached to a first wafer. One or more other package components may then be attached and connected to the first package component, and the structure may then be flipped for subsequent processing (e.g., formation of conductive connectors). For example, the first carrier is removed, and the dielectric bond layer of the first package component is planarized. The planarization may reach protrusions of the aluminum pads, which may result in a smearing of the protrusions (e.g., a conductive material) over the dielectric bond layer. As such, an etch process may be utilized to remove the smeared portion of the protrusion from atop the dielectric bond layer as well as etching down any exposed protrusions. Dielectric layers may then be formed over the dielectric bond layer, and conductive connectors may be formed through those dielectric layers for connection to others of the aluminum pads. The removal of the smeared portion and reduction in the height of the protrusion improve reliability and yield of the semiconductor package during subsequent processing, which can reduce manufacturing costs. In addition, the removal and reduction will improve reliability of the completed semiconductor package during functional use.

[0008] Embodiments discussed herein are to provide examples to enable making or using the subject matter of this disclosure, and a person having ordinary skill in the art will readily understand modifications that can be made while remaining within contemplated scopes of different embodiments. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. Although method embodiments may be discussed as being performed in a particular order, other method embodiments may be performed in any logical order.

[0009] FIGS. 1-5, 6A-7B, and 8-12 illustrate the cross-sectional views and perspective views of intermediate stages in the fabrication process of a semiconductor package, including wafer acceptance testing (WAT) and process control monitoring (PCM) processes as well as subsequent steps in the fabrication process to further increase efficiency and improve quality.

[0010] In FIG. 1, a first package component **40** is formed, for example, in a wafer (not separately illustrated). In accordance with some embodiments, first package components **40** are individual

device dies, packages having one or more device dies packaged therein, System-on-Chip (SoC) dies including a plurality of integrated circuits (or device dies) integrated as a system, or the like. The device die(s) in first package components **40** may be or may comprise logic dies, memory dies, input-output dies, Integrated Passive Devices (IPDs), the like, or combinations thereof. For example, the logic device die(s) in first package components **40** may be Central Processing Unit (CPU) dies, Graphic Processing Unit (GPU) dies, mobile application dies, Micro Control Unit (MCU) dies, BaseBand (BB) dies, Application processor (AP) dies, or the like. The memory die(s) in first package components **40** may include Static Random Access Memory (SRAM) dies, Dynamic Random Access Memory (DRAM) dies, or the like. The device die(s) in first package components **40** may include semiconductor substrates and interconnect structures.

[0011] In accordance with some embodiments, first package component **40** may include a substrate **41** comprising a semiconductor substrate **42** (e.g., a silicon substrate), integrated circuit devices (not separately illustrated) at a front-side surface of semiconductor substrate **42**, a plurality of dielectric layers **44** formed over semiconductor substrate **42** and the integrated circuit devices. The integrated circuit devices may include active devices, passive devices, and the like. Substrate **41** of first package component **40** further includes through-silicon vias (TSVs) **46** and an interconnect structure **48**. In some embodiments, the interconnect structure **48** may include a seal ring, for example, such as an outer perimeter of conductive elements embedded in the plurality of dielectric layers **44**. For example, the outer perimeter of these conductive elements may be electrically disconnected from the rest of the interconnect structure **48** and the integrated circuit devices. TSVs **46** may extend partially through semiconductor substrate **42** as well as partially through plurality of dielectric layers **44**.

[0012] Interconnect structure **48** is disposed over the front-side of the semiconductor substrate **42** and embedded in plurality of dielectric layers **44**. Interconnect structure **48** includes metal lines and vias electrically connected to the integrated circuit devices. As illustrated, interconnect structure **48** includes a plurality of levels of the metal lines, wherein one or more levels of upper metal lines **48U** may be coupled with corresponding ones of TSVs **46**.

[0013] Metal pads **50/52**, including aluminum pads **50** and sacrificial pads **52**, are disposed over substrate **41** (e.g., over plurality of dielectric layers **44**) and electrically connected to upper metal lines **48U** of interconnect structure **48** by conductive vias **54**. Aluminum pads **50** will help provide external electrical connection during functional use of first package components **40**, and sacrificial pads **52** will provide external electrical connection during circuit probe testing of first package components (see FIG. 2). Aluminum pads **50** and sacrificial pads **52** may have the same compositions, such as comprising aluminum, an aluminum-copper alloy (e.g., comprising up to 1% copper by volume), or any suitable material. Aluminum pads **50** and sacrificial pads **52** may each have heights $H_{\text{sub.1}}$ over substrate **41** in a range from 1.2 μm to 3.0 μm . Metal pads **50/52** may be coated with a dielectric layer **56** for protection, such as from oxidation. For example, the dielectric layer **56** may be an anti-reflective coating (ARC) and comprise an oxide or a nitride, such as silicon oxynitride (SiON), silicon carbide (SiC), or any suitable material.

[0014] In some embodiments, aluminum pads **50** and sacrificial pads **52** (e.g., metal pads **50/52**) are formed simultaneously over substrate **41**. For example, a sacrificial material (not shown) may first be formed over substrate **41**. Openings are formed in the sacrificial material by first applying a photoresist over a top surface of the sacrificial material and patterned using a photolithographic mask. The patterned photoresist is then used as an etching mask to etch openings in the sacrificial material and expose portions of substrate **41** (e.g., conductive vias **54**). To form the openings, the sacrificial material may be etched by a suitable process such as dry etching (e.g., reactive ion etching (RIE) or neutral beam etching (NBE), etc.), wet etching, or the like. In other embodiments, the sacrificial material itself is the photoresist, and an energy source (e.g., ultraviolet light) is shined through the photomask to change chemical properties (e.g., solubility) of regions of the sacrificial material impinged by the energy. To form the openings, those regions of the sacrificial

material may be etched by a suitable process such as an isotropic wet etch process.

[0015] The openings within the sacrificial material are filled with a conductive material to form pads **50/52**. In an embodiment, the conductive material may comprise a seed layer and a plate metal (not separately illustrated). The seed layer may be blanket deposited over the exposed top surfaces of substrate **41**, and may comprise, for example, a copper layer. The seed layer may be deposited using processes such as sputtering, evaporation, or plasma-enhanced chemical vapor deposition (PECVD), or the like, depending upon the desired materials. The plate metal may be deposited over the seed layer through a plating process such as electrical or electro-less plating. The plate metal may comprise aluminum, an aluminum-copper alloy, or the like. In some embodiments, the sacrificial material and pads **50/52** may be planarized, for example, using a chemical mechanical polish (CMP) or a grinding process. The photoresist, the sacrificial material, and portions of the seed layer on which the conductive material is not formed are removed. The photoresist and/or the sacrificial material may be removed by one or more acceptable ashing or stripping processes, such as using an oxygen plasma or the like. Afterward, exposed portions of the seed layer are removed, such as by using an acceptable etching process. The remaining portions of the conductive material (e.g., the seed layer and the plate metal) form pads **50/52**.

[0016] Although only one aluminum pad **50** and one sacrificial pad **52** are illustrated, it should be appreciated that each first package component **40** includes a plurality of each type of metal pad **50/52**. Some aluminum pads **50** may be connected to TSVs **46** by upper metal lines **48U**. Some aluminum pads **50** may be connected to the integrated circuit devices at the surface of semiconductor substrate **42** by interconnect structure **48**. In addition, some sacrificial pads **52** may be connected to corresponding aluminum pads **50** that are connected to the integrated circuit devices through upper metal lines **48U** of interconnect structure **48**.

[0017] In FIG. **2**, a first wafer acceptance testing (WAT) processing, such as circuit probe testing, is performed on first package components **40** to ascertain whether first package components **40** are known good dies (KGDs). First package components **40** may be tested using one or more probes **62**. Probes **62** are physically and electrically coupled to sacrificial pads **52** by, e.g., reflowable test connectors. Although not specifically illustrated, more than one probe **62** (e.g., two probes **62**) may be coupled to corresponding sacrificial pads **52** (e.g., two sacrificial pads **52**). Only wafers with first package components **40** which are KGDs undergo subsequent processing and packaging, and wafers with first package components **40** which fail the circuit probe testing are not subsequently processed and packaged. The testing may include providing power, ground, and signal voltages to sacrificial pads **52** in order to test the functionality of the various first package components **40** (e.g., the integrated circuit devices and interconnect structure **48** within). In some embodiments, the circuit probe testing may include testing for known open or short circuits that may be expected based on the design of the integrated circuits within the first package components **40**. In some embodiments, after testing is complete, probe **62** is removed and any excess reflowable material on the sacrificial pads **52** may be removed by, e.g., an etching process, a chemical-mechanical polish (CMP), a grinding process, or the like.

[0018] As discussed above, ones of sacrificial pads **52** undergoing the circuit probe testing may be electrically connected through interconnect structure **48** to the integrated circuit devices located at the semiconductor substrate **42**. Upper metal lines **48U** of interconnect structure may electrically connect ones of aluminum pads **50** to these ones of sacrificial pads **52**. As such, these ones of sacrificial pads **52** are electrically connected to the integrated circuit devices for the circuit probe testing, while these ones of aluminum pads **50** are similarly electrically connected to the integrated circuit devices for the functional use of the completed semiconductor package.

[0019] As illustrated, the circuit probe testing may result in deformation of sacrificial pads **52**, for example, forming probe marks in sacrificial pads **52**. For example, probe marks may include a recess **52R** (e.g., caused by physical contact from probe **62**) and a protrusion **52P** (e.g., caused by displacement of material of sacrificial pad **52** by the physical contact from probe **62**) in the

corresponding sacrificial pad **52**. Consequently, recess **52R** may have a depth D.sub.1 of up to 1.1 μm . In addition, protrusion **52P** indicates vertical displacement of material of sacrificial pad **52** to a height H.sub.2 above a top major surface of sacrificial pad **52** being up to 1.2 μm , while being less than height H.sub.1. In addition, the circuit probe testing may result in portions of dielectric layer **56** located at recess **52R** and protrusion **52P** of sacrificial pad **52** being displaced or removed, while dielectric layer **56** remains over other portions of sacrificial pad **52** and over aluminum pad **50**.

[0020] In FIG. 3, a dielectric bond layer **58** is formed over aluminum pad **50** and sacrificial pad **52** of first package component **40**. Dielectric bond layer **58** may be a single homogenous layer or a composite of two or more layers comprising, for example, an oxide and/or a nitride, such as silicon oxide (SiO.sub.2), silicon oxynitride (SiON), silicon nitride (SiN), the like, or any suitable material(s). Dielectric bond layer **58** may be formed using spin coating, Flowable Chemical Vapor Deposition (FCVD), or the like. Dielectric bond layer **58** is formed to have a top surface above an entirety of protrusion **52P** of sacrificial pad **52**. In accordance with some embodiments, after forming dielectric bond layer **58**, individual first package components **40** are singulated from the wafer, using any suitable sawing process, in order for the KGDs of first package components **40** to undergo subsequent processing and packaging as discussed below.

[0021] In FIG. 4, singulated first package components **40** are bonded to a first carrier **20** through an oxide-oxide bonding process, such as fusion bonding. Although one first package component **40** is illustrated, there may be a plurality of first package components **40** (e.g., KGDs) bonded to first carrier **20**. The plurality of first package components **40** may be discrete package components physically separate from each other, and the bonding process is a die-to-wafer bonding. In embodiments not illustrated in which first package components **40** are not first singulated, first package components **40** may be within the un-sawed wafer, and the bonding process is a wafer-to-wafer bonding.

[0022] First carrier **20** includes a base carrier **22** and one or more dielectric bond layers **24**. In some embodiments, base carrier **22** may be a wafer, and may be formed of a same material as semiconductor substrate **42** in the overlying first package component **40**, so that in the subsequent packaging process, warpage caused by mismatch of Coefficients of Thermal Expansion (CTE) is reduced. For example, base carrier **22** may be formed of or comprise silicon, while other materials such as laminate, ceramic, glass, silicate glass, or the like, may also be used. In accordance with some embodiments, the entire base carrier **22** is formed of a homogeneous material, with no other material different from the homogeneous material therein. For example, the entire base carrier **22** may be formed of silicon (doped or undoped), and without a metal region, dielectric region, etc., therein.

[0023] Before attaching first package components **40** to first carrier **20**, dielectric bond layers **24** are deposited on base carrier **22**. Dielectric bond layers **24** may include oxide based materials (which may also be silicon oxide based) such as silicon oxide (SiO), phospho-silicate glass (PSG), borosilicate glass (BSG), boron-doped phospho silicate glass (BPSG), fluorine-doped silicate glass (FSG), nitride-based materials such as silicon nitride (SiN), oxynitride based materials such as silicon oxynitride (SiON), while it may also be formed of or comprises other materials such as silicon oxycarbide (SiOC), silicon carbonitride (SiCN), or the like. Dielectric bond layers **24** may be formed using spin coating, FCVD, Plasma Enhanced Chemical Vapor Deposition (PECVD), Low Pressure Chemical Vapor Deposition (LPCVD), Atomic Layer Deposition (ALD), the like, or combinations thereof. For example, in some embodiments, dielectric bond layers **24** may include a lowermost layer (e.g., proximal to base carrier **22**) comprising an oxide, one or more middle layers comprising a nitride and/or an oxynitride, and an uppermost layer (e.g., distal from base carrier **22**) comprising an oxynitride (e.g., with a lower nitrogen-to-oxygen ratio as compared with the middle layers). Although not specifically illustrated, alignment marks may be formed in dielectric bonding layers **24** (e.g., the uppermost layer) using any suitable method.

[0024] In accordance with some embodiments, the bonding of first package components **40** to first

carrier **20** includes pre-treating dielectric bond layers **24** and **58** in a process gas comprising oxygen (O.sub.2) and/or nitrogen (N.sub.2), performing a pre-bonding process to join dielectric bond layers **24** and **58** together, and performing an annealing process following the pre-bonding process. In accordance with some embodiments, during the pre-bonding process, first package components **40** are put into contact with first carrier **20**, with a pressing force applied to press first package components **40** against first carrier **20**. The pre-bonding may be performed at room temperature (in a range from 20° C. to 25° C.), and a higher temperature may also be used.

[0025] After the pre-bonding, an annealing process is performed. Chemical bonds, such as Si—O—Si bonds, may be formed between dielectric bond layers **24** and **58**, so that dielectric bond layers **24** and **58** are bonded to each other with high bonding strength. In accordance with some embodiments, the annealing process is performed at a temperature in a range from 200° C. to 400° C. The annealing duration may be in a range from 30 minutes to 60 minutes.

[0026] In FIG. 5, after attaching first package components **40** to first carrier **20**, liner layer **30** and gap-filling material **32** are formed over first package components **40** and first carrier **20** to encapsulate first package components **40**. Liner layer **30** may be a conformal layer extending along the top surfaces and the sidewalls of first package components **40** and along top surfaces of dielectric bond layer **24**. Liner layer **30** may also be referred to as a moisture stop layer and, in some embodiments, used as an etch stop layer in subsequent steps. Liner layer **30** may be formed of a dielectric material that has good adhesion to the sidewalls of first package components **40**. For example, liner layer **30** may be formed of an extra low-k (ELK) material, including a nitride such as silicon nitride and/or an oxide such as silicon oxide. The deposition of liner layer **30** may include a conformal deposition process such as ALD, CVD, or any suitable process. Gap-filling material **32** may be formed of a molding compound, an epoxy, a resin, and/or the like. For example, gap-filling material **32** may comprise a nitride such as silicon nitride and/or an oxide such as silicon oxide and may be deposited using spin coating, FCVD, PECVD, LPCVD, ALD, or any suitable process.

[0027] A planarization process such as a CMP process and/or a mechanical grinding process is then performed to remove portions of liner layer **30** and gap-filling material **32** from over the back-side surface (the illustrated top surface) of first package components **40**. The planarization process may continue in order to thin portions of semiconductor substrate **42** until TSVs **46** are exposed. After the planarization process, semiconductor substrate **42** may be coplanar (within process variations) with top surfaces of liner layer **30** and gap-filling material **32**.

[0028] In FIGS. 6A and 6B, dielectric bond layer **34** and bond pads **35** are formed over the back-side surface of first package component **40** (e.g., the back-side surface of semiconductor substrate **42**). In some embodiments, dielectric bond layer **34** may be first deposited using any suitable method such as ALD, CVD, or the like. Dielectric bond layer **34** is then patterned to form openings, which are filled with a conductive material to form bond pads **35**, similarly as described above in connection with metal pads **50/52**.

[0029] FIG. 6B illustrates an embodiment in which dielectric bond layer **34** and bond pads **35** are formed over semiconductor substrate **42** and TSVs **46** before forming liner layer **30** and gap-filling material **32**. In some such embodiments, dielectric bond layer **34** and bond pads **35** may be formed before singulating and attaching first package components **40** to first carrier **20** (see FIG. 4). As such, dielectric bond layer **34** and bond pads **35** may be considered part of first package components **40**. After forming liner layer **30** and gap-filling material **32**, a planarization process such as a CMP process and/or a mechanical grinding process is performed to remove portions of liner layer **30** and gap-filling material **32** from over the back-side surface of first package components **40**. The planarization process may continue until bond pads **35** and dielectric bond layer **34** are exposed. As illustrated, after performing the planarization process, liner layer **30** and gap-filling material **32** may be coplanar (within process variations) with dielectric bond layer **34** and bond pads **35**.

[0030] In FIGS. 7A-7B, second package components **70** are attached to first package components

40. FIG. 7A illustrates the attachment to the structure illustrated in FIG. 6A, and FIG. 7B illustrates the attachment to the structure illustrated in FIG. 6B. Notably, subsequent figures illustrate the subsequent processing steps performed on the structure of FIG. 7A, however, a person of ordinary skill in the art would understand how those subsequent processing steps would also apply to the structure of FIG. 7B.

[0031] Second package components **70** may include second package components **70A** comprising integrated circuits and second package components **70B** serving as dummy package components. For example, second package components **70A** may be attached through a hybrid bonding process, and second package components **70B** may be attached through an oxide-oxide bonding process, such as fusion bonding. Although one of second package components **70A** and one of second package components **70B** are illustrated, there may be a plurality of second package components **70A** and/or a plurality of second package components **70B** attached to first package component **40**. The plurality of second package components **70A/B** may be discrete package components physically separate from each other (e.g., singulated from their respective wafers).

[0032] In accordance with some embodiments, second package components **70A** may be the same as, similar to, or different from first package components **40**. For example, second package components **70A** are individual device dies, packages having one or more device dies packaged therein, System-on-Chip (SoC) dies including a plurality of integrated circuits (or device dies) integrated as a system, or the like. The device die(s) in second package components **70A** may be or may comprise logic dies, memory dies, input-output dies, Integrated Passive Devices (IPDs), the like, or combinations thereof. For example, the logic device die(s) in second package components **70A** may be Central Processing Unit (CPU) dies, Graphic Processing Unit (GPU) dies, mobile application dies, Micro Control Unit (MCU) dies, BaseBand (BB) dies, Application processor (AP) dies, or the like. The memory die(s) in second package components **70A** may include Static Random Access Memory (SRAM) dies, Dynamic Random Access Memory (DRAM) dies, or the like. The device die(s) in second package components **70A** may include semiconductor substrates and interconnect structures.

[0033] In accordance with some embodiments not specifically illustrated, second package components **70A** may include features similar to those described above in first package components **40**. For example, second package components **70A** may include a semiconductor substrate, integrated circuit devices, and a plurality of dielectric layers formed over the semiconductor substrate and the integrated circuit devices. The integrated circuit devices may include active devices, passive devices, and the like. Second package components **70A** may further include dielectric bond layer **76** with bond pads **77** embedded within.

[0034] In accordance with some embodiments, second package components **70B** may be dummy package components that do not contribute functional integrated circuits and/or are electrically disconnected from first package components **40** and second package components **70A**. Second package components **70A** may provide structural support for the semiconductor package as well as heat dissipation from first package components **40** and/or second package components **70A** during functional use of the semiconductor package.

[0035] Still referring to FIGS. 7A-7B, in some embodiments, before attaching second package components **70A/B** to first package components **40**, a dielectric bond layer **36** and bond pads **37** may be formed over dielectric bond layer **34** and bond pads **35**. Dielectric bond layer **36** and bond pads **37** may be formed similarly as described above in connection with dielectric bond layer **34** and bond pads **35**, respectively.

[0036] The bonding of second package components **70A** to first package components **40** may be achieved through hybrid bonding, in which both of metal-to-metal direct bonding (between bond pads **37** and **77** or between bond pads **37** and **35** if bond pads **37** are formed on second package component **70A** before bonding) and dielectric-to-dielectric bonding (such as Si—O—Si bonding between surface dielectric bond layers **36** and **76** or between dielectric bond layers **36** and **34** if

dielectric bond layer **36** is formed on second package component **70A** before bonding) are formed. Furthermore, there may be a single or a plurality of second package components **70A** bonded to the same first package component **40**. The plurality of second package components **70A** bonded to the same first package component **40** may be identical to, or different from, each other.

[0037] In accordance with some embodiments, dielectric bond layer **76** is bonded to the dielectric bond layer **36** through dielectric-to-dielectric bonding, without using any adhesive material (e.g., die attach film). Similarly, the bond pads **77** are bonded to bond pads **37** through metal-to-metal bonding, without using any eutectic material (e.g., solder). The bonding may include a pre-bonding and an annealing. During the pre-bonding, a small pressing force is applied to press second package components **70A** against first package components **40**. The pre-bonding is performed at a low temperature, such as room temperature, such as a temperature in the range from 15° C. to 30° C., and after the pre-bonding, the dielectric bond layer **76** and dielectric bond layer **36** are bonded to each other. The bonding strength is then improved in a subsequent annealing step, in which dielectric bond layer **76** and dielectric bond layer **36** are annealed at a high temperature, such as a temperature in the range from 100° C. to 450° C. After the annealing, bonds, such as fusion bonds, are formed bonding dielectric bond layer **76** with dielectric bond layer **36**. For example, the bonds can be covalent bonds between the material of dielectric bond layer **76** and the material of dielectric bond layer **36**. Bond pads **77** and bond pads **37** are connected to each other with a one-to-one correspondence. Bond pads **77** and bond pads **37** may be in physical contact after the pre-bonding, or may expand to be brought into physical contact during the annealing. Further, during the annealing, the material of bond pads **77** (e.g., copper) and the material of bond pads **37** (e.g., copper) intermingles, so that metal-to-metal bonds are also formed. Hence, the resulting bonds between second package components **70A** and first package components **40** are hybrid bonds that include both dielectric-to-dielectric bonds and metal-to-metal bonds.

[0038] The bonding of second package components **70B** to first package components **40** may be performed before, after, or at various points during attachment of second package components **70A**. The bonding of second package components **70B** is achieved through oxide-oxide bonding, in which dielectric-to-dielectric bonding (such as Si—O—Si bonding between surface dielectric bond layers **36** and **78**) is formed. Furthermore, there may be a single or a plurality of second package components **70B** bonded to the same first package component **40**. The plurality of second package components **70B** bonded to the same first package component **40** may be identical to, or different from, each other.

[0039] After attaching second package components **70A/B** to first package components **40**, a liner layer **80** and a gap-filling material **82** are formed over and between second package components **70A/B** to encapsulate second package components **70A/B**. Liner layer **80** may be a conformal layer extending along the top surfaces and the sidewalls of second package components **70** as well as along top surfaces of dielectric bond layer **36** and may serve as a moisture stop layer. Liner layer **80** is formed of a dielectric material that has good adhesion to the sidewalls of second package components **70**. For example, liner layer **80** may be formed of an extra low-k (ELK) material, including a nitride such as silicon nitride and/or an oxide such as silicon oxide. The deposition of liner layer **80** may include a conformal deposition process such as ALD, CVD, or any suitable process. Gap-filling material **82** may be formed of a molding compound, an epoxy, a resin, and/or the like. For example, gap-filling material **82** may comprise a nitride such as silicon nitride and/or an oxide such as silicon oxide. For example, liner layer **80** and gap-filling material **82** may be formed similarly as described above in connection with liner layer **30** and gap-filling material **32**, respectively.

[0040] Liner layer **80** and gap-filling material **82** may be planarized to become level with top surfaces of second package components **70**. For example, a thinning process may be utilized, such as a CMP process, a grinding process, an etch back process, the like, or combinations thereof.

[0041] In FIG. **8**, a second carrier **90** is bonded to second package components **70** through an

oxide-oxide bonding process, such as fusion bonding. Second carrier **90** includes a base carrier **92** and one or more dielectric bond layers **94**. Base carrier **92** may be a wafer, and may be formed of a same material as, for example, base carrier **22** in first carrier **20**, so that in the subsequent packaging process, the warpage caused by mismatch of Coefficients of Thermal Expansion (CTE) is reduced. In accordance with some embodiments, base carrier **92** may be formed of or comprise silicon, while other materials such as laminate, ceramic, glass, silicate glass, or the like, may also be used. In accordance with some embodiments, the entire base carrier **92** is formed of a homogeneous material, with no other material different from the homogeneous material therein. For example, the entire base carrier **92** may be formed of silicon (doped or undoped), and there is no metal region, dielectric region, etc., therein.

[0042] Before attaching second carrier **90** to second package components **70**, one or more dielectric bond layers **94** are deposited on base carrier **92** and one or more dielectric bond layers **86** are deposited over second package components **70**. Dielectric bond layers **86** and **94** may be formed similarly as described above in connection with any of dielectric bond layers **24/34/36/76**.

[0043] In accordance with some embodiments, the bonding of second carrier **90** to second package components **70** includes pre-treating dielectric bond layers **86** and **94** in a process gas comprising oxygen (O.sub.2) and/or nitrogen (N.sub.2), performing a pre-bonding process to join dielectric bond layers **86** and **94** together, and performing an annealing process following the pre-bonding process. In accordance with some embodiments, during the pre-bonding process, second carrier **90** is put into contact with dielectric bond layers **86**, with a pressing force applied to press second carrier **90** against second package components **70**. The pre-bonding may be performed at room temperature (in a range from 20° C. to 25° C.), and a higher temperature may also be used.

[0044] After the pre-bonding, an annealing process is performed. Chemical bonds, such as Si—O—Si bonds, may be formed between dielectric bond layers **86** and **94**, so that dielectric bond layers **86** and **94** are bonded to each other with high bonding strength. In accordance with some embodiments, the annealing process is performed at a temperature in a range from 200° C. to 400° C. The annealing duration may be in a range from 30 minutes to 60 minutes.

[0045] In FIG. **9**, the structure may be flipped, first carrier **20** is removed, and dielectric bond layer **58** is exposed. First carrier **20** may be removed using any suitable method, such as a planarization process. The planarization process may continue until dielectric bond layer **58** is coplanar (within process variations) with gap-filling material **32** and liner layer **30**. The planarization process may be a CMP process, a grinding process, the like, or a combination thereof. In accordance with some embodiments, the planarization process may form a smeared region **95** due to an upper portion of protrusion **52P** smearing over and/or along an upper surface of dielectric bond layer **58**. As further illustrated, following the planarization process, a new top surface of protrusion **52P** may be level with dielectric bond layer **58**, gap-filling material **32**, and liner layer **30**.

[0046] In FIG. **10**, a wet etch process is performed to remove smeared region **95** that was formerly the upper portion of protrusion **52P**. The wet etch process may be performed using etchants having a high selectivity toward etching the material of sacrificial pad **52** (e.g., aluminum and/or an aluminum-copper alloy) the materials of dielectric bond layer **58**, gap-filling material **32**, and liner layer **30** remain substantially non-etched and intact. For example, the etch process may be an isotropic wet etch process comprising etchants such as HF, H.sub.2SO.sub.4/H.sub.2O.sub.2, H.sub.2O, and combinations thereof. The wet etch process may be performed at standard temperature and pressure for a duration in a range from 15 seconds to 3 minutes. In addition to removing smeared region **95**, the wet etch process may also remove a new upper portion of protrusion **52P**, thereby forming recess **96** from the top surface of dielectric bond layer **58**, wherein recess **96** is directly over protrusion **52P** of sacrificial pad **52**.

[0047] As discussed above, sacrificial pads **52** have top major surfaces with height H.sub.1 above substrate **41**, and sacrificial pads **52** may have recesses **52R** formed during the circuit probe testing (see FIG. **2**) with depth D.sub.1 measured from the top major surfaces of sacrificial pads **52**.

However, due to the step of planarizing dielectric bond layer **58** and protrusion **52P** (see FIG. **9**) followed by the wet etch process performed on protrusion **52P** and smeared region **95**, protrusion **52P** may now have a height $H_{sub.3}$ above the top major surfaces of sacrificial pads **52**. For example, height $H_{sub.3}$ is less than previous height $H_{sub.2}$ (as well as height $H_{sub.1}$ of sacrificial pads **52**), such as in a range from 0.2 μm to 1.1 μm . In some embodiments, height $H_{sub.3}$ of protrusion **52P** after etching may be in a range from 6.6% to 91% of height $H_{sub.1}$ of protrusion **52P** before etching.

[0048] As further illustrated, an uppermost surface of protrusion **52P** may be located at a depth $D_{sub.2}$ below the top surface of dielectric bond layer **58**. By comparison, the top major surfaces of sacrificial pads **52** may be located at a depth $D_{sub.3}$ below a top surface of dielectric bond layer **58**. For example, depth $D_{sub.2}$ of protrusion **52P** below the top surface of dielectric bond layer **58** may be in a range from 0.1 μm to 1.0 μm . Further, depth $D_{sub.3}$ of top surfaces of sacrificial pads **52** below the top surface of dielectric bond layer **58** may be in a range from 0.4 μm to 1.2 μm .

[0049] In FIG. **11**, a passivation layer **102** and a dielectric layer **104** are formed over aluminum pads **50** and sacrificial pads **52** (e.g., metal pads **50/52**). In addition, openings **106** are formed through passivation layer **102**, dielectric layer **104**, dielectric bond layer **58**, and dielectric layer **56** to expose aluminum pads **50**. Passivation layer **102** may be a single layer or a plurality of layers and be conformally deposited as an oxide such as silicon oxide, a nitride such as silicon nitride, silicon oxynitride, the like, or combinations thereof. For example, a silicon oxide lower layer (not separately illustrated) may be first deposited over the dielectric bond layer **58** and fill an entirety of recess **96**. A silicon nitride upper layer (also not separately illustrated) may then be deposited over the silicon oxide lower layer. The single or plurality of layers comprising passivation layer **102** may be deposited using any suitable method, such as CVD, ALD, combinations thereof, or any suitable methods. Passivation layer **102** forms over dielectric bond layer **58** with improved efficiency and with fewer deficiencies after previously removing smeared region **95**. The improved deposition results in a stronger bond between passivation layer **102** and dielectric bond layer **58**, thereby reducing risk of delamination during subsequent processing steps.

[0050] Passivation layer **102** may be formed to have an average thickness in a range from 2.0 μm to 3.5 μm . In some embodiments, passivation layer **102** (e.g., conformally deposited) may have a top surface containing a recess **102R** directly above protrusion **52P** of sacrificial pad **52** (e.g., directly above the location of recess **96** formed in connection with FIG. **10**). As a result, recess **102R** in the top surface of passivation layer may have depth $D_{sub.4}$ less than depth $D_{sub.2}$ (e.g., depth $D_{sub.4}$ is shallower than depth $D_{sub.2}$), such as in a range from 100 nm to 900 nm. In other embodiments, the top surface of passivation layer **102** may be as level near the portion directly above **102R** as in other portions elsewhere.

[0051] Passivation layer **102** may then be patterned to form openings **106** through passivation layer **102** and through dielectric bond layer **58**. For example, a photoresist may be applied over a top surface of passivation layer **102** and patterned. The patterned photoresist is then used as an etching mask to etch portions of passivation layer **102** in order to form openings **106** and expose dielectric layer **56**. Passivation layer **102** may be etched by a suitable process such as dry etching (e.g., reactive ion etching (RIE) or neutral beam etching (NBE), etc.), wet etching, or the like. In some embodiments, the etchants may be selected such that passivation layer **102** and/or dielectric bond layer **58** have high etch selectivities as compared with that for dielectric layer **56**, and dielectric layer **56** therefore serves as an etch stop layer during the etching process.

[0052] Dielectric layer **104** may be formed of a polymer material such as polyimide, PBO, the like, or any suitable material. Dielectric layer **104** may be conformally deposited using, for example, spin coating, FCVD, PECVD, LPCVD, ALD, the like, or combinations thereof. As illustrated, dielectric layer **104** is formed over the top surface and along exposed sidewalls of passivation layer **102** as well as along exposed sidewalls of dielectric bond layer **58** within openings **106**. As such, dielectric layer **104** partially fills and extends entirely across openings **106**. A top surface of

dielectric layer **104** may be substantially planar, including in a location directly above recess **102R** (if present) of passivation layer **102** and directly above protrusion **52P** of sacrificial pad **52**. [0053] Dielectric layer **104** may then be patterned to re-form openings **106**. For example, a photoresist may be applied over a top surface of dielectric layer **104** and patterned. The patterned photoresist is then used as an etching mask to etch dielectric layer **104** and dielectric layer **56** (if still present in openings **106**) in order to re-form openings **106** and expose aluminum pads **50**. Dielectric layer **104** may be etched by a suitable process such as dry etching (e.g., RIE or NBE, etc.), wet etching, or the like. In embodiments in which dielectric layer **56** was used as an etch stop layer in the prior patterning process, dielectric layer **56** may be etched in the same or a subsequent etching process to expose aluminum pads **50**. This patterning process may be performed similarly as described above in connection with the prior patterning process, albeit using etchants selected to etch both dielectric layers **56** and **104**.

[0054] In FIG. **12**, under-bump metallurgies (UBMs) **110** and conductive connectors **124** are formed over and through dielectric layer **104** and passivation layer **102** to be electrically connected to aluminum pads **50**. As an example to form UBMs **110**, a seed layer (not separately illustrated) is formed over the exposed surfaces of aluminum pads **50** and dielectric layer **104**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. In some embodiments, the seed layer includes a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to UBMs **110**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is then formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may include a metal, such as copper, titanium, tungsten, aluminum, or the like. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process. The remaining portions of the seed layer and conductive material form UBMs **110**.

[0055] Conductive connectors **124** are then formed on UBMs **110**. Conductive connectors **124** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. Conductive connectors **124** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, conductive connectors **124** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into desired bump shapes. In another embodiment, conductive connectors **124** comprise metal pillars (such as copper pillars) formed by sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder-free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

[0056] Although not specifically illustrated, the semiconductor package may undergo a second wafer acceptance testing (WAT) process or multiple WAT processes. The second WAT processes may involve subjecting the semiconductor package to extreme conditions (e.g., high temperature, humidity, and/or pressure) while testing functionality of the integrated circuits within. For example,

the highly accelerated temperature and humidity stress test (HAST) allows for an accelerated testing of longevity and reliability by being conducted in extreme environmental conditions.

[0057] Advantages of the disclosed embodiments may be achieved and, for example, realized during the second WAT processes. In particular, removal of smeared region **95** and/or reducing protrusion **52P** before forming UBMs **110** and conductive connectors **124** improve performance of the integrated circuits by minimizing or preventing undesired short circuits and parasitic capacitance. The stronger bonds between passivation layer **102** and dielectric bond layer **58** (e.g., due to removal of smeared region **95**) may also improve performance and reliability of the semiconductor package during the second WAT processes as well as functional use.

[0058] Although also not specifically illustrated, the semiconductor package may be singulated by sawing through second carrier **90**, dielectric layer **104**, and layers therebetween. The singulation may be performed before, after, or in between the second WAT processes. Further advantages of the disclosed embodiments may be achieved and, for example, realized during singulation of the semiconductor package. In particular, the stronger bonds between passivation layer **102** and dielectric bond layer **58** due to removal of smeared region **95**, which would otherwise be interposed between the layers. As a result, delamination of passivation layer **102** from dielectric bond layer **58** is prevented or reduced during singulation of the semiconductor package. Indeed, delamination is similarly prevented or reduced during any other processing steps that may tend to cause warpage to parts or all of the semiconductor package.

[0059] In an embodiment, a method includes attaching a first package component to a first carrier, the first package component comprising: an aluminum pad disposed adjacent to a substrate; a sacrificial pad disposed adjacent to the substrate, the sacrificial pad comprising a major surface opposite the substrate, a protrusion of the sacrificial pad extending from the major surface; and a dielectric bond layer disposed around the aluminum pad and the sacrificial pad; attaching a second carrier to the first package component and the first carrier, the first package component being interposed between the first carrier and the second carrier; removing the first carrier; planarizing the dielectric bond layer to comprise a top surface being coplanar with the protrusion; and etching a portion of the protrusion. In another embodiment, the method further includes, before attaching the first package component: performing a circuit probe test on the first package component, the performing the circuit probe test comprising forming the protrusion of the sacrificial pad; and forming the dielectric bond layer over the protrusion of the sacrificial pad. In another embodiment, the performing the circuit probe test further comprises: electrically connecting a power signal to the sacrificial pad; and forming a first recess in the major surface of the sacrificial pad. In another embodiment, the method further includes, before attaching the second carrier, hybrid bonding a second package component to the first package component. In another embodiment, the method further includes, before attaching the second carrier, oxide-oxide bonding a dummy package component to the first package component. In another embodiment, after etching the portion of the protrusion, the protrusion is recessed below the top surface of the dielectric bond layer. In another embodiment, the method further includes forming a passivation layer over the dielectric bond layer, wherein a portion of the passivation layer directly above the protrusion comprises a second recess. In another embodiment, the method further includes: forming a dielectric layer over the passivation layer; and forming an opening through the dielectric layer, the passivation layer, and the dielectric bond layer to expose the aluminum pad.

[0060] In an embodiment, a method includes performing a circuit probe test on a first package component, the first package component comprising an aluminum pad and a sacrificial pad disposed over a substrate, the performing the circuit probe test comprising: physically coupling a probe to the sacrificial pad; and connecting a power or ground signal to the sacrificial pad, wherein after connecting the power or ground signal to the sacrificial pad, the sacrificial pad comprises a protrusion extending upward from a first top surface of the sacrificial pad; forming a first dielectric bond layer over the substrate, the aluminum pad, and the sacrificial pad of the first package

component, the first dielectric bond layer comprising a second top surface opposite the substrate; planarizing the first dielectric bond layer, the planarizing exposing the protrusion of the sacrificial pad; and performing a wet etch process to remove an upper portion of the protrusion. In another embodiment, the substrate of the first package component comprises: an interconnect structure electrically connected to the sacrificial pad; a through-silicon via electrically connected to the aluminum pad; and a semiconductor substrate, the through-silicon via extending partially through the semiconductor substrate. In another embodiment, the method further includes: attaching the second top surface of the first dielectric bond layer to a first carrier; thinning the semiconductor substrate to expose the through-silicon via; and forming a second dielectric bond layer and a bond pad adjacent to the first package component, the through-silicon via being electrically interposed between the bond pad and the aluminum pad. In another embodiment, the method further includes hybrid bonding a second package component to the first package component using the second dielectric bond layer and the bond pad. In another embodiment, after connecting the power or ground signal to the sacrificial pad, the sacrificial pad comprises a recess extending downward from the first top surface of the sacrificial pad. In another embodiment, the sacrificial pad comprises aluminum, and wherein performing the wet etch process comprises using etchants being highly selective to etch aluminum over material of the first dielectric bond layer.

[0061] In an embodiment, a semiconductor package includes: a semiconductor substrate disposed over a package component; a through-silicon via extending through an entirety of the semiconductor substrate, the through-silicon via being electrically connected to the package component; an interconnect structure disposed over the semiconductor substrate, the interconnect structure electrically connected to integrated circuit devices at a surface of the semiconductor substrate; a first aluminum pad disposed over the semiconductor substrate and electrically connected to the interconnect structure, the first aluminum pad comprising: a top major surface opposite the semiconductor substrate; a first recess extending toward the semiconductor substrate; and a protrusion extending away from the semiconductor substrate; a second aluminum pad disposed over the semiconductor substrate and electrically connected to the through-silicon via; a first dielectric bond layer disposed over and around the first aluminum pad and the second aluminum pad; and a passivation layer disposed over the first aluminum pad and in physical contact with the protrusion. In another embodiment, the semiconductor package further includes a third aluminum pad disposed over the semiconductor substrate and electrically connected to the first aluminum pad and to the interconnect structure. In another embodiment, the semiconductor package further includes an under-bump metallurgy disposed over and in physical contact with the second aluminum pad. In another embodiment, the semiconductor package further includes: a first bonding layer disposed along a bottom surface of the semiconductor substrate; a first bond pad embedded in the first bond layer; a second bonding layer disposed along an upper surface of the package component; and a second bond pad embedded in the second bond layer, the second bond pad physically contacting the first bond pad. In another embodiment, a height of the protrusion above the top major surface is less than a height of the top major surface from a lower major surface. In another embodiment, the passivation layer comprises a second recess directly above the protrusion.

[0062] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: forming a package component, forming the package component comprising: forming an integrated circuit over a semiconductor substrate; forming a first metal pad and a second metal pad over the integrated circuit; performing a circuit probe test on the first metal pad comprising forming a first recess and a protrusion in the first metal pad; depositing a first dielectric bond layer over the first metal pad and the second metal pad; bonding the first dielectric bond layer to a carrier; attaching an integrated circuit die to the semiconductor substrate; removing the carrier; removing a portion of the first dielectric bond layer and a first portion of the protrusion; after removing the portion of the first dielectric bond layer and the first portion of the protrusion, removing a second portion of the protrusion; after removing the portion of the first dielectric bond layer, forming a passivation layer over the first dielectric bond layer; and forming an under-bump metallurgy through the passivation layer and electrically connected to the second metal pad.
2. The method of claim 1, wherein a height of the protrusion is greater than a depth of the first recess of the first metal pad.
3. The method of claim 1, wherein removing the portion of the first dielectric bond layer and the first portion of the protrusion comprises smearing a material of the protrusion along a surface of the first dielectric bond layer.
4. The method of claim 1, wherein removing the second portion of the protrusion comprises performing a wet etch process.
5. The method of claim 4, wherein etchants of the wet etch process comprise at least one of hydrogen fluoride, sulfuric acid, or hydrogen peroxide.
6. The method of claim 4, wherein after removing the portion of the first dielectric bond layer and the first portion of the protrusion, the first portion of the protrusion is level with a surface of the first dielectric bond layer, and wherein after removing the second portion of the protrusion, the protrusion is recessed below the surface of the dielectric bond layer.
7. The method of claim 6, wherein a top surface of the passivation layer comprises a second recess directly above a remainder of the protrusion, and wherein a depth of the protrusion below the surface of the dielectric bond layer is greater than a depth of the second recess.
8. The method of claim 1, wherein the first metal pad comprises aluminum.
9. A method comprising: forming a first integrated circuit die in a wafer, the first integrated circuit die comprising: an integrated circuit over a substrate; a first aluminum pad and over and electrically connected to the integrated circuit, a first top surface of the first aluminum pad comprising a first recess extending toward the integrated circuit and a protrusion extending away from the integrated circuit; a second aluminum pad over and electrically connected to the integrated circuit, a second top surface of the second aluminum pad being substantially flat; and a first dielectric layer over and around the first aluminum pad and the second aluminum pad; singulating the first integrated circuit die from the wafer; bonding the first dielectric layer of the first integrated circuit die to a second dielectric layer along a first carrier substrate; attaching a second integrated circuit die over the substrate of the first integrated circuit die; attaching a second carrier substrate over the second integrated circuit die; removing the first carrier substrate; removing the second dielectric layer; removing a portion of the first dielectric layer to expose the protrusion; performing an etch process with an etchant that is selective for aluminum over a material of the first dielectric layer; and after performing the etch process, forming a third dielectric layer over the first dielectric layer and the protrusion.
10. The method of claim 9, wherein removing the portion of the first dielectric layer to expose the protrusion comprises forming a smear region over the first dielectric layer, wherein the smear region comprises aluminum from the protrusion.
11. The method of claim 10, wherein performing the etch process comprises removing the smear

region from the first dielectric layer.

- 12.** The method of claim 10, wherein performing the etch process comprises a second recess of the protrusion from an upper surface of the first dielectric layer.
 - 13.** The method of claim 9, wherein forming the first integrated circuit die comprises performing a circuit probe test on the first aluminum pad, wherein performing the circuit probe test comprises forming the first recess and the protrusion.
 - 14.** The method of claim 13, wherein before performing the circuit probe test, an oxidation layer is over the first top surface of the first aluminum pad, and wherein after performing the circuit probe test, the first recess and the protrusion are free of the oxidation layer.
 - 15.** A method comprising: forming active devices and an interconnect structure over a semiconductor substrate; forming a first metal pad and a second metal pad over the interconnect structure; forming a first dielectric protective layer over the first metal pad and a second dielectric protective layer over the second metal pad; contacting the first metal pad with a test probe, after contacting the first metal pad the first metal pad comprising a recess at a point of contact with the test probe and a protrusion adjacent to the recess; depositing a first dielectric layer over the first metal pad and the second metal pad; performing a planarization process on the first dielectric layer to expose the protrusion; performing an etch process to remove a material of the protrusion; depositing a second dielectric layer over the first dielectric layer; patterning an opening in the second dielectric layer to expose the second metal pad; and forming an under-bump metallurgy in the opening over the second metal pad.
 - 16.** The method of claim 15, wherein after contacting the first metal pad with the test probe, the recess and the protrusion are free of the first dielectric protective layer.
 - 17.** The method of claim 16, wherein after patterning the opening in the second dielectric layer, an exposed portion of the second metal pad is free of the second dielectric protective layer.
 - 18.** The method of claim 15, wherein the etch process comprises hydrogen fluoride or a combination of sulfuric acid and hydrogen peroxide.
 - 19.** The method of claim 15, wherein a height of the protrusion is greater than a depth of the recess.
 - 20.** The method of claim 15, further comprising: attaching the first dielectric layer to a first carrier; planarizing the semiconductor substrate to expose a through substrate via; attaching an integrated circuit die over and electrically connected to the through substrate via; attaching a second carrier to the integrated circuit die; and removing the first carrier.
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